



A QUICK ENGINEERING BREATH

ONE WRONG NUMBER, FOUR INHERITED MISTAKES

Everything downstream trusts the process description it was given.

if this is wrong ↓



this inherits the mistake

Ku and Pu aren't measured from the real process — they're computed from whatever gain, lag and delay you told the model to assume.

Get that description wrong, and the "correct" tuning it produces is confidently wrong.

What that actually looks like — next.



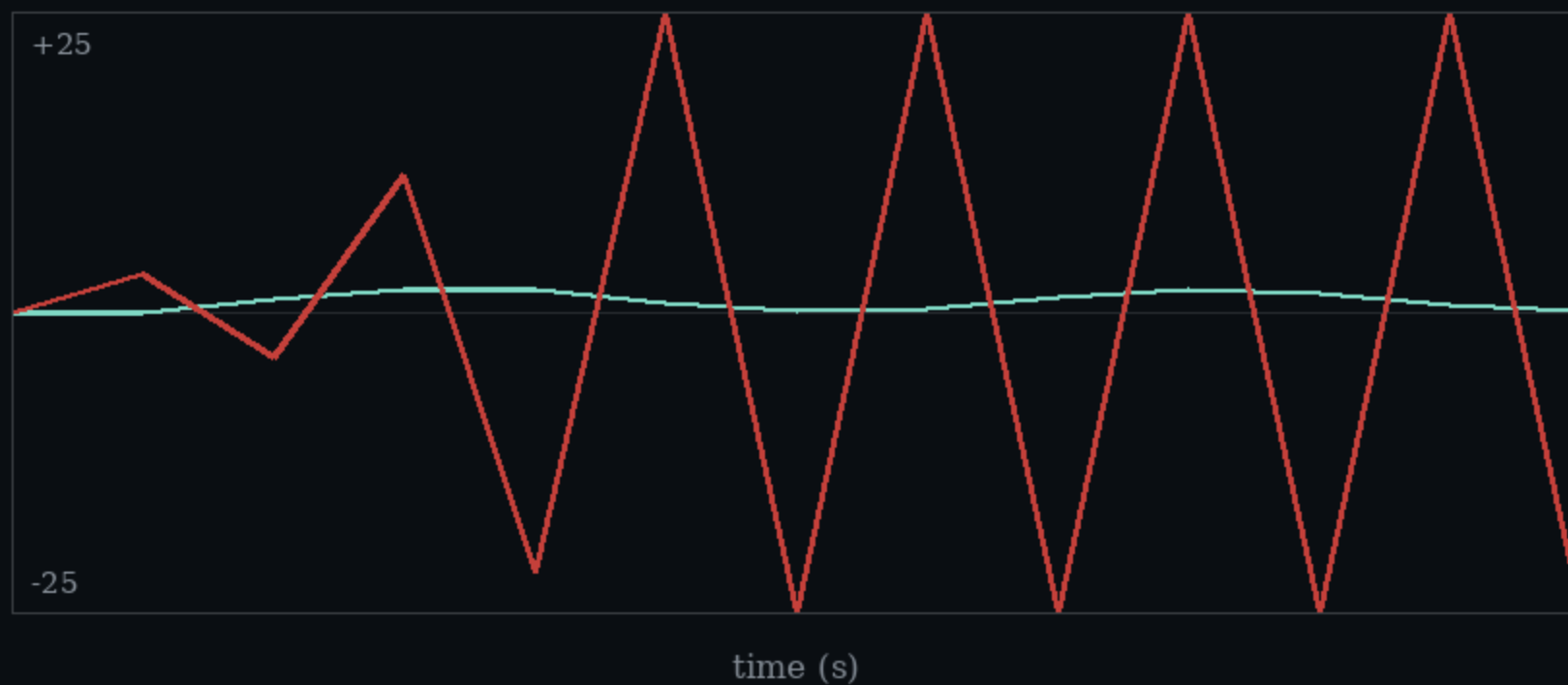
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SAME PROCESS, THE WRONG MODEL

Assumed delay: 0.3s. Real delay: 1s. Same tuning method — tuned on the wrong number.

— correctly identified (delay = 1s)

— misidentified (assumed delay = 0.3s)



The red curve doesn't stop at the edge of this chart — it keeps growing. By $t=20s$ it's in the millions.

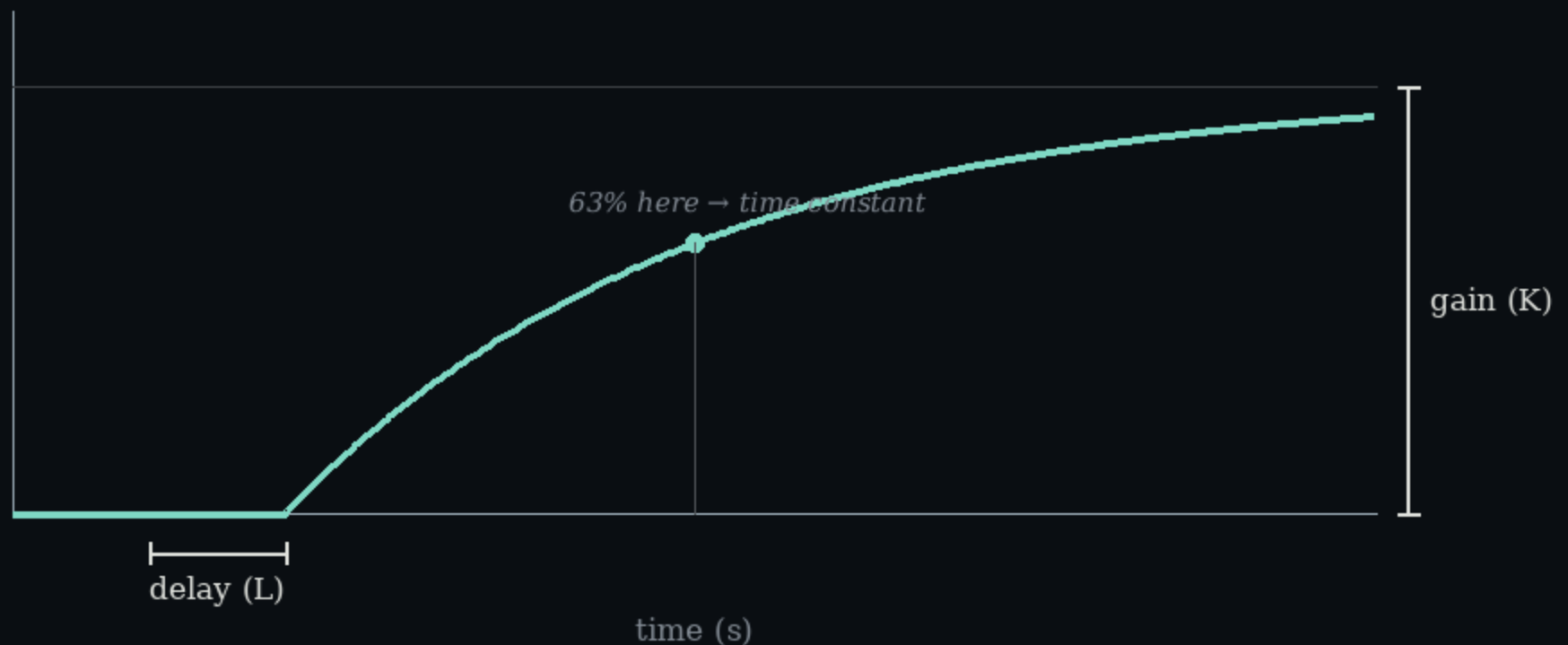
The tuning was never wrong. The process description fed into it was.



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MEASURE IT. DON'T ASSUME IT.

The safe way to get gain, lag and delay: bump the real process once and read the curve.



Three numbers off one honest test — no guessing, no risk to the real loop.